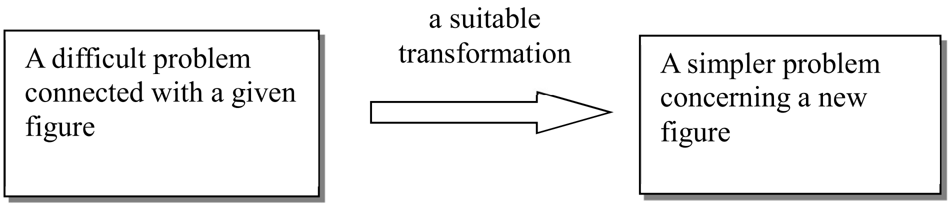
Olympiad Geometry – IV

1. Homothety

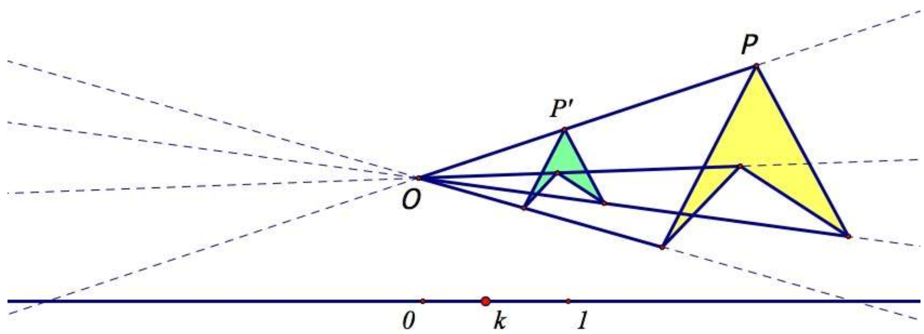
One of the most useful methods exploited by geometers of the modern era is that of cleverly transforming a figure into another which is better suited to a geometrical investigation.



Definition
Let O be a fixed point of the plane and k a given nonzero real number. By the homothety (or expansion, or dilatation, or stretch) $H(O, k)$ we mean the transformation of the plane onto itself, which maps each point P of the plane into the point P' of the plane such that:

(a) Points O, P and P' are collinear.

(b) $\overrightarrow{OP'} = k \cdot \overrightarrow{OP}$

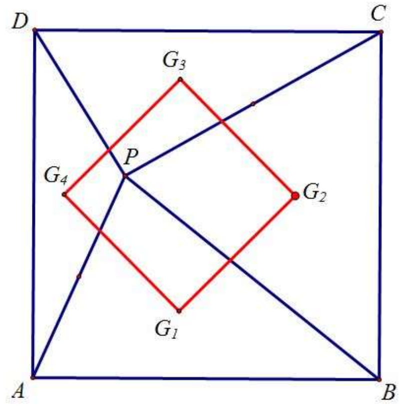


The point O is called the centre of the homothety, and k is called the ratio of the homothety.
If $k > 0$, the homothety is a magnification with centre O. If $k < 0$, it is a reduction with centre O.
Observe that choice $k = -1$ corresponds to point reflection.

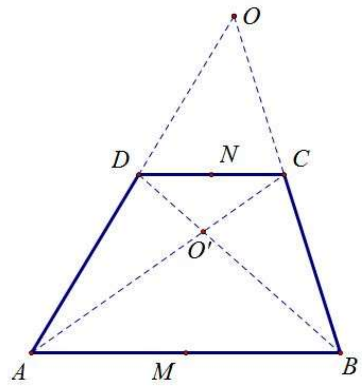
Basic Properties of Homothety
Given a homothety $H(O, k)$, the following properties will apply:

1. H maps O to O and every point M different from O to a point M' on the line OM.
2. H maps every line L into a line parallel to L; if L passes through O then H maps L into itself.
3. If the points A, B and X lies on a line, A', B' and X' are their image under H, then $\dfrac{A'X'}{X'B'} = \dfrac{AX}{XB}$.
4. Let the images of distinct non-collinear points A, B and C be A', B' and C' respectively. Then
   (a) $A'B' // AB$ and $A'B' = kAB$.
   (b) (**Homothety preserves angles and ratios**) $\angle A'B'C' = \angle ABC$ and $\dfrac{A'B'}{B'C'} = \dfrac{AB}{BC}$.
5. H maps any polygon M into another polygon which is similar to M and whose sides are parallel to the respectively sides of M. In fact, H maps any figure into a figure similar in shape.
6. (**Correspondence Principle for Homothety**) The image of every element (point, line, segment, etc) of any figure F is a correspondence element of the image F' of F, which is located in a similar way and has similar properties to F, e.g. H maps a median of a triangle into the respective median of the image of this triangle, its incircle into the incircle of the image, and so on.

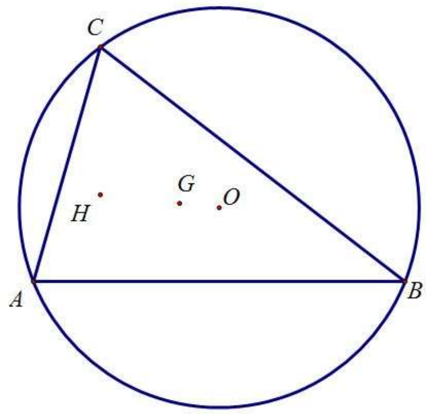
Example 1.1 (Tournament of Towns 1984)
Let P be a point inside a given square *ABCD*. Prove that the centroids of triangles *ABP*, *BCP*, *CDP*, *DAP* form a square.



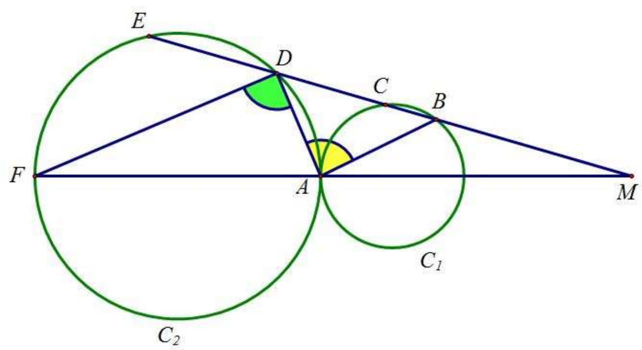
Example 1.2
Prove that in every trapezium, the midpoints of the parallel sides, the intersection point of the diagonals, and the intersection point of the non-parallel sides produced are collinear.



Example 1.3 (Euler line)
Let *ABC* be a non-equilateral triangle and let *H*, *G*, *O* be its orthocentre, centroid, and circumcentre respectively. Then the points *H*, *G*, *O* lie on a single line (called the **Euler line** of triangle *ABC*) in this order, and $HG = 2GO$.

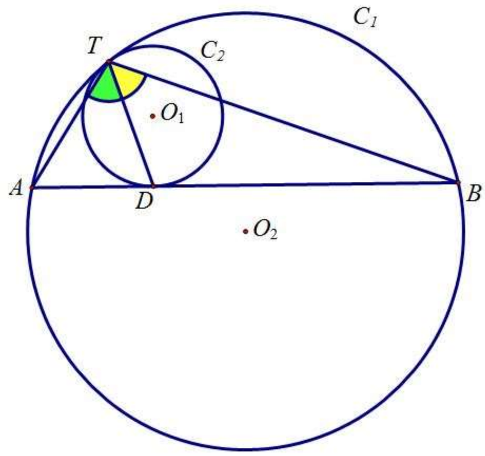


Example 1.4
Two circles are tangent, externally, at point *A*. A line passing through the intersection point of their common external tangents intersects the two circles at the point *B*, *C*, *D* and *E* in succession. Prove that $\angle BAD = 90^\circ$.



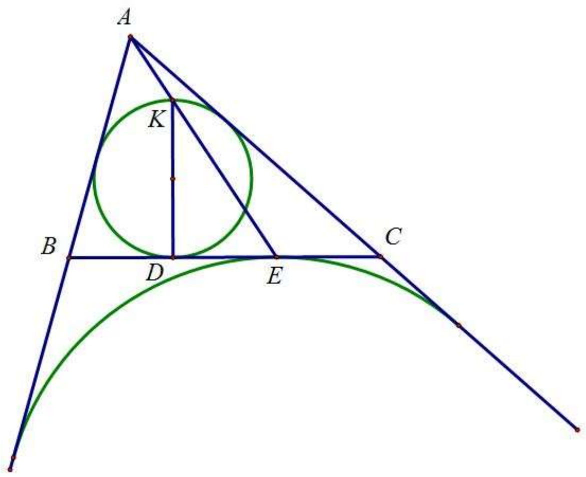
Example 1.5

Two circles touch each other internally at the point T. The chord AB in the larger circle touches the smaller circle at point D. Show that the line TD bisects $\angle ATB$.



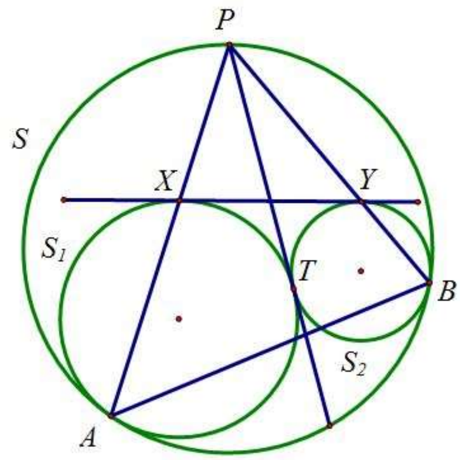
Example 1.6

Let ABC be a triangle and let its incircle and the A-excircle touch the side BC at D, E respectively. Let K be the point on the incircle such that KD is a diameter. Then A, K, E lie on a single line.



Example 1.7

Circles S_1 and S_2 are internally tangent to a circle S at distinct points A, B respectively. Moreover, they are tangent to each other at T. Denote by P any intersection of S and their common tangent through T. Let the lines PA, PB intersect S_1 and S_2 for the second time at X, Y respectively. Show that XY is a common tangent of S_1 and S_2.



Example 1.8 (IMO 2006 SL-G2)

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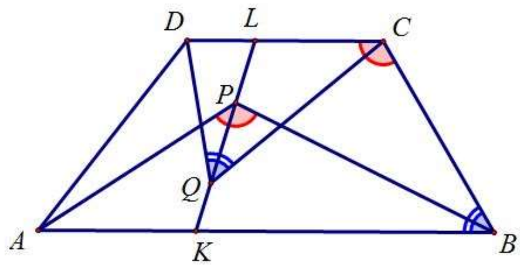
Let $ABCD$ be a trapezoid with parallel sides $AB > CD$. Points K and L lie on the line segments AB and CD, respectively, so that $\dfrac{AK}{KB} = \dfrac{DL}{LC}$. Suppose that there are points P and Q on the line segment KL satisfying $\angle APB = \angle BCD$ and $\angle CQD = \angle ABC$.

Prove that the points P, Q, B and C are concyclic.



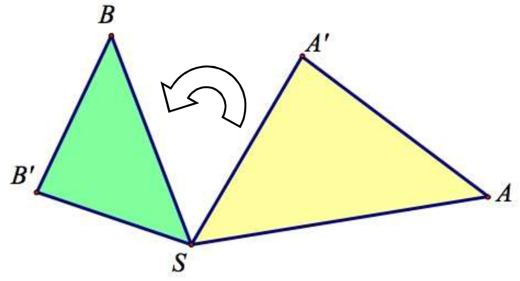
2. Spiral Similarity

Given a point S, a positive number k, and an angle φ different from 0° and 180°, **spiral similarity** with centre S, dilation factor k, and angle of rotation φ is a geometric transformation that sends point A to a point A' such that:

(a) $SA' = k\,SA$,

(b) $\angle(SA, SA') = \varphi$

Such a spiral similarity is denoted by $\mathsf{S}(S, k, \varphi)$.



If we allowed $\varphi = 0°$ or $\varphi = 180°$, spiral similarity would reduce to homothety. For $k = 1$ it reduces to rotation. In general, spiral similarity is a composition of these two transformations.

Basic Properties of Spiral Similarity

Let $\mathsf{S}(S, k, \varphi)$ be a spiral similarity that maps A to A' and B to B' etc. Then:

1. Image of a line L is a line. If we denote it by L', then $\angle(L, L') = \varphi$.
2. Image of a triangle ABC is a triangle $A'B'C'$ directly similar to it with factor k. In other words,

$$\frac{A'B'}{AB} = \frac{A'C'}{AC} = \frac{B'C'}{BC} = k$$

   and $\angle(AB, A'B') = \angle(AC, A'C') = \angle(BC, B'C') = \varphi$.

3. Image of a circle with radius R is a circle with radius kR.

One thing to remember about spiral similarities is that they come in pairs. Whenever we come across a spiral similarity, there is always another one nearby.

4. (a) $\Delta SAB \sim \Delta SA'B'$

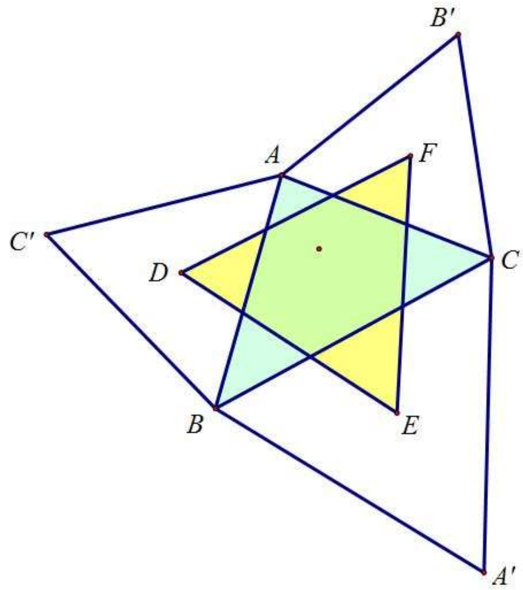
   (b) $\Delta SAA' \sim \Delta SBB'$

   (c) Spiral similarity $\mathsf{S}'(S, k', \varphi')$ maps A to B and A' and B' for suitable choice of k' and φ'.

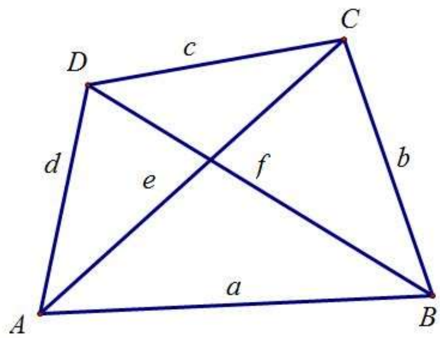
Example 2.1

Let A, B, A', B' be four distinct points in the plane, such that no three of them are collinear. is not parallel. Assume that the lines AB and $A'B'$ meet at P. Then there exists unique spiral similarity that sends A to A' and B to B'. The centre of this spiral similarity is the second intersection of the circumcircles of triangles $AA'P$ and $BB'P$.

Example 2.2 (Napoleon's Theorem)
Let ABC be a triangle and let BCD, CAE, ABF be equilateral triangles erected outwards from its sides. Show that the centroids A_1, B_1, C_1 of these equilateral triangles also form an equilateral triangle.

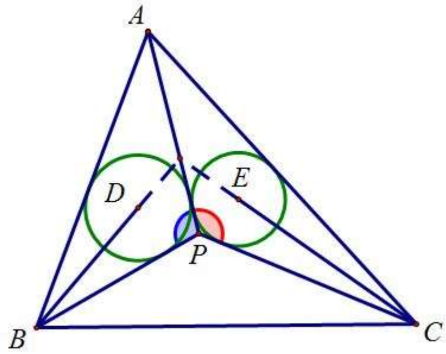


Example 2.3 (Ptolemy's Inequality)
Let $ABCD$ be a quadrilateral. Denote the lengths of AB, BC, CD, DA by a, b, c, d respectively, and its diagonals AC, BD by e, f respectively.
Then $ac + bd \geq ef$.



Example 2.4 (IMO 1996)

Let P be a point inside a triangle ABC such that $\angle APB - \angle ACB = \angle APC - \angle ABC$.
Let D, E be the incentres of triangles APB, APC, respectively.
Show that the lines AP, BD, CE meet a point.



Exercise (IMO 2019 Hong Kong Team Selection Test 2 (October))

Let Γ_1 and Γ_2 be two circles with different radii, with Γ_1 the smaller one. The two circles meet at distinct points A and B. C and D are two points on the circle Γ_1 and Γ_2 respectively, and such that A is the midpoint of the segment CD. CB is extended to meet the circle Γ_2 at F, while DB is extended to meet the circle Γ_1 at E. The perpendicular bisectors of CD and EF meet at P.

(a) Prove that $\angle EPF = 2\angle CAE$

(b) Prove that $AP^2 = CA^2 + PE^2$.